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Alleviation of Fufang Fanshiliu decoction on type II diabetes mellitus by reducing insulin resistance: A comprehensive network prediction and experimental validation

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ABSTRACT

Ethnopharmacological relevance: Fufang Fanshiliu decoction (FFSLD) is a Chinese herbal medicine prescription that has been used in type 2 diabetes mellitus (T2DM), while the underlying mechanism remains unclear.
Aim of the study: To validate the efficacy and explore the potential mechanisms of FFSLD in treating T2DM via integrating a network pharmacological approach and experimental evaluation.
Materials and methods: T2DM mice model induced by high-fat diet feeding combined with streptozotocin injection was selected to investigate the alleviation of FFSLD against T2DM, via detecting the levels of glucose, insulin, glucagon (GC), triglyceride (TG), total cholesterol (TC), high-density lipoprotein cholesterol (HDL-C), and low-density lipoprotein cholesterol (LDL-C). Network pharmacological analysis was used to predict the potential mechanisms, including the pharmacokinetics and drug-likeness screening, active ingredients and potential targets prediction, network analysis, and enrichment analysis. The candidate bioactive molecules of FFSLD, and targets information excavated through TCMSP, Uniprot, GeneCards, OMIM databases, were combined for comprehensive analysis by constructing "drug-compound-target-disease" and "protein-protein interaction" networks. Enrichment analysis was performed via Gene Ontology (GO) and Koto Encyclopedia of Genes and Genomes (KEGG) databases. HepG2 insulin-resistance (IR) cells model induced by high glucose was used to verify the potential mechanisms of FFSLD against T2DM which were predicted by the network pharmacology.
Results: The animal study showed that FFSLD significantly decreased the blood glucose, and reversed the abnormal levels of insulin, GC, TG, TC, HDL-C, and LDL-C in T2DM mice. Network pharmacological analysis indicated that 106 active compounds of FFSLD might be correlated with 628 targets in treating T2DM, and the mechanism would probably be related to insulin resistance that harbored a high response value ($P = 5.88844$ E^{-33}) though regulating Akt1, ESR1, oxidoreductase activity, and JAK/STAT signalings. Experimental validation showed that FFSLD reduced the ROS level, up-regulated the expressions of p-AKT, Nrf-2, and ESR1, and down-regulated the expressions of JAK2, STAT3, and Keap-1 in the HepG2-IR cells model.
Conclusions: This study demonstrated that the therapeutic effect of FFSLD on T2DM was related to IR alleviation. The underlying mechanisms were associated with the regulation of PI3K/AKT, JAK/STAT, oxidative stress, and ESR signaling pathways.

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1. Introduction

Diabetes mellitus is a kind of metabolic syndrome which seriously endangers the health of human beings, and its incidence has increased rapidly in recent decades (Cheema et al., 2014; Walls et al., 2014; Domingueti et al., 2016). According to the latest data released by the International Diabetes Federation (IDF), about 463 million adults aged 20 ~ 79 years in the world suffering from diabetes, of which 90% belonged to type II diabetes mellitus (T2DM), and the number would be reaching 578 million by 2030 according to IDF forecasting (Saeedi et al., 2019). Chinese herbal medicine, such as Sanggua drink (Cai et al., 2018), Tianmai Xiaoke tablet (Wang et al., 2016), has a long history and good efficacy in diabetes

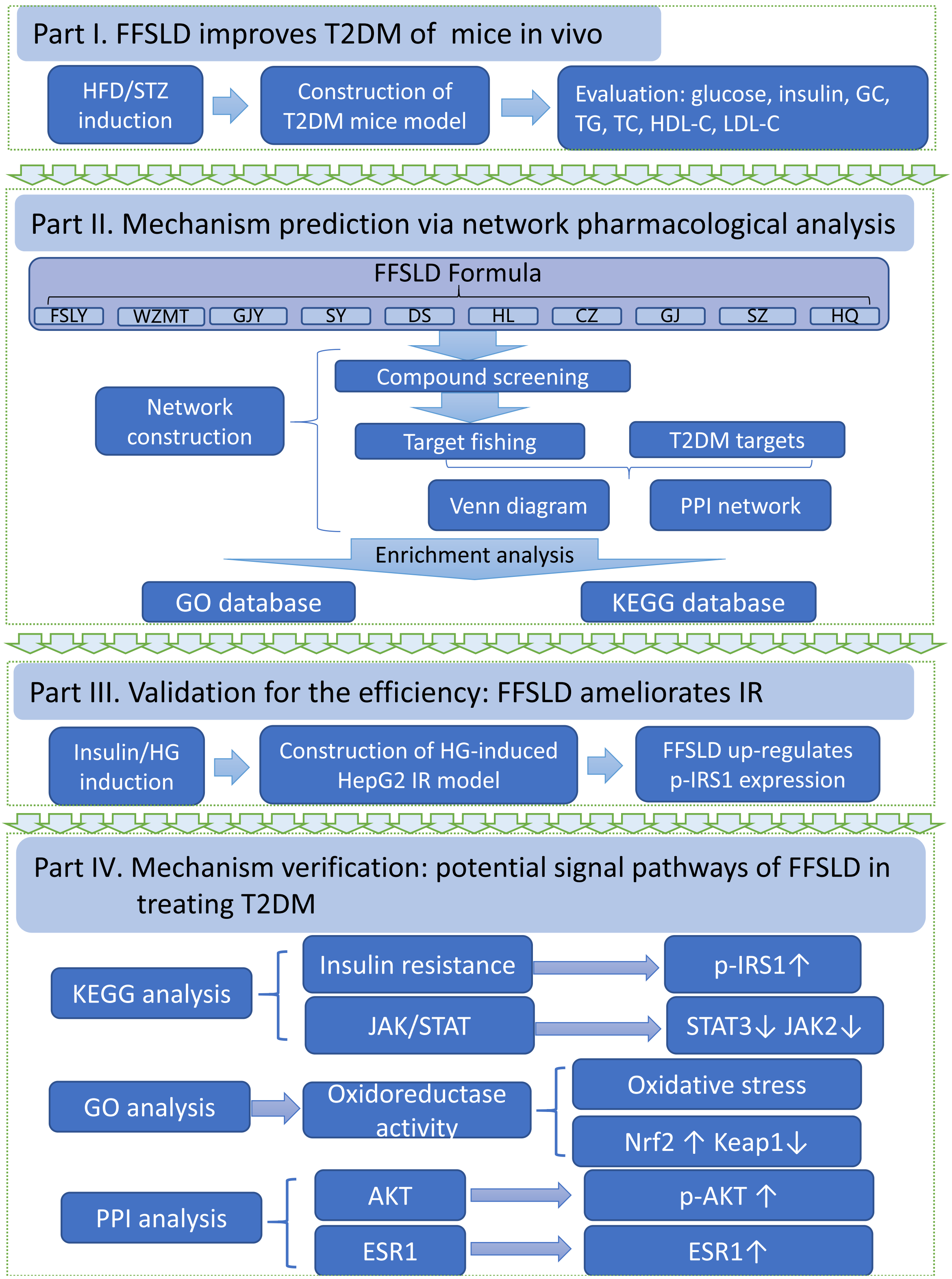

Fig. 1. Workflow for the study of FFSLD in anti-type II diabetes mellitus (T2DM) and -insulin resistance (IR) effects based on integrating network pharmacological approach and experimental evaluation.

therapy which offered a bright prospect for clinical application.

Fufang Fanshiliu decoction (FFSLD) is a traditional Chinese herbal medicine prescription composed of *Psidium guajava* L. (Fanshiliuye), *Ficus simplicissima* Lour. (Wuzhimaotao), *Euonymus alatus (Thunb.)* Sied. (Guijianyu), *Morus alba* L. (Sangye), *Codonopsis pilosula* N. (Dangshen), *Coptis chinensis* F. (Huanglian), *Atractylodes lancea* DC. (Cangzhu), *Zingiber officinale* R. (Ganjiang), *Crataegus pinnatifida* B. (Shanzha), and *Monascus purpureus* W. (Hongqu). The clinical effects of FFSLD have been verified in resolving phlegm, decreasing fasting blood glucose, and ameliorating abnormal lipid metabolism (Lin et al., 2014, 2016; Li et al., 2015, 2016; Cheng et al., 2016). Our previous studies also showed that FFSLD inhibited the activities of α-amylase and α-glucosidase (Dai et al., 2016). Furthermore, our newly granted patent demonstrated that FFSLD effectively improved the symptoms of patients with T2DM (Li et al., 2019). However, the underlying pharmacological mechanisms remain undefined.

Network pharmacology is a novel strategy to predict the potential targets of drugs widely used in recent decades (Zhao et al., 2015; Huang et al., 2020), especially in the study of multi-components of Chinese herbal medicine (Xu et al., 2020). The "drug-compound-disease-target" network and "protein to protein interaction" (PPI) network are constructed based on the prediction of active multi-components of drugs and therapeutic targets of diseases (Lyu et al., 2017). The result of the network analysis would provide a direction for the following study of the detailed pharmacological mechanism and signaling pathways. Insulin resistance (IR) is a major symptom and an important cause leading to T2DM, of which several signaling pathways abnormally regulated are involved, such as PI3K/Akt, PKC, MAPKs, AMPK, NF-κB, JAK/STAT, oxidative stress, and ESR (Wu et al., 2006; Szendroedi et al., 2014; Luo et al., 2015; Wang et al., 2016; Kalpana et al., 2019; Lu et al., 2021).

This study was designed to investigate the efficacy of FFSLD in treating T2DM in mice and explore the underlying mechanism via integrating network pharmacological methodology and experimental validation. In this study, the T2DM mice model induced by high-fat diet (HFD) feeding combined with streptozotocin (STZ) injection was used to evaluate the therapeutic effect of FFSLD against T2DM *in vivo* through detecting the levels of glucose, insulin, glucagon (GC), triglyceride (TG), total cholesterol (TC), high-density lipoprotein cholesterol (HDL-C), and low-density lipoprotein cholesterol (LDL-C). Network pharmacology was employed to predict the potential mechanism of FFSLD, by which the high-response targets and signaling pathways were selected out. Furthermore, the HepG2-IR cells model induced by high glucose (HG) (Ding et al., 2019) was used to validate the IR alleviation of FFSLD *in vitro*, and Western Blot, flow cytometry methodologies were used to verify the potential mechanism and signaling pathways that predicted by the network pharmacological approach. The design of the whole research is shown in Fig. 1.

2. Materials and methods

2.1. Chemicals and reagents

Dulbecco's Modified Eagle Medium (DMEM), fetal bovine serum (FBS), penicillin-streptomycin, and phosphate-buffered saline (PBS) were purchased from Gibco (MA, USA). ROS staining kit, bovine serum albumin (BSA), and 3-[4, 5-dimethylthiazol-2-yl]-2, 5-diphenyltetrazolium bromide (MTT) were purchased from Beyotime Biotechnology (Shanghai, China). Primary antibodies against p-IRS1 Ser307 (#AF3272), Nrf-2 (#AF0639), Keap-1 (#AF5266), SREBP-1c (#AF6283), and estrogen receptor-1 (#AF6058) were purchased from Affinity Biosciences (OH, USA). Primary antibodies against β-actin (#12262), p-AKT (#4060), and secondary antibodies (anti-mouse and anti-rabbit) were purchased from Cell Signaling Technology (Boston, MA, USA). Streptozocin (#S0130) was from Sigma-Aldrich Co., Ltd. (St. Louis, MO, USA).

2.2. Drug preparation

FFSLD was composed of *Psidium guajava* L. (Fanshiliuye), *Ficus simplicissima* Lour. (Wuzhimaotao), *Morus alba* L. (Sangye), *Codonopsis pilosula* N. (Dangshen), *Coptis chinensis* F. (Huanglian), *Atractylodes lancea* DC. (Cangzhu) and *Crataegus pinnatifida* B. (Shanzha), *Euonymus alatus* (Thunb.) Sied. (Guijianyu), *Monascus purpureus* W. (Hognqu) and *Zingiber officinale* R. (Ganjiang) at the ratio of 6:4:3:6:2:3:3:3:3:2. The materials of *Psidium guajava* L. (#C22103062), *Ficus simplicissima* Lour. (#C22108042), *Morus alba* L. (#C22110151), *Codonopsis pilosula* N. (#C12108207), *Coptis chinensis* F. (#C22105098), *Atractylodes lancea* DC. (#C22105060) and *Crataegus pinnatifida* B. (#C22106013) were got from Sinopharm Feng Liaoxing (Foshan) Herbal Pieces Co., Ltd (Guangdong, China), *Euonymus alatus* (Thunb.) Sied. (#200601) and *Monascus purpureus* W. (#210901) were purchased from Zhixin Chinese pharmaceutical Co., Ltd (Guangdong, China), and *Zingiber officinale* R. (#210801) was bought from Bencaotang Chinese pharmaceutical Co., Ltd (Guangxi, China). FFSLD was prepared by Zhongshan Hospital of Traditional Chinese Medicine (Zhongshan, China). The materials of FFSLD were moistened, boiled for 2 h, and the extraction was collected. After that, the extraction was evaporated, freeze-dried to obtain the powder, and then stored at −20. The extraction yield of FFSLD was 15%. The powder was reconstituted with normal saline at the indicated concentrations for the *in vivo* study, and dissolved in DMEM medium, followed by sterilization using a membrane filter (0.22 μm) for the *in vitro* study.

2.3. Cell line and culture

HepG2 cells were purchased from Cell Bank of Shanghai Institute of Cell Biology, Chinese Academy of Sciences (Shanghai, China). Cells were cultured in DMEM containing 10% FBS and 1% penicillin-streptomycin and maintained in an incubator with 5% CO_2 equilibrated at 37 °C.

2.4. MTT assay

In brief, cells were seeded in 96-well plates, and treated with different concentrations of FFSLD (0, 125, 250, 500, 1000, 2000, 4000 μg/mL). After treatment for 24 h, cells were added with 20 μL of MTT solution (5 mg/mL) and incubated for 4 h in dark at 37 °C. The supernatant was discarded, and then 150 μL DMSO was added to dissolve the formazan. The absorbance of each well was measured at 570 nm using a Multiskan™ FC Microplate Photometer (Thermo Scientific, Shanghai, China).

2.5. Effect of FFSLD on HepG2-IR cells model induced by HG

Cells in the logarithmical growth phase were stimulated by insulin (1 μM) for 0–60 min, and then proteins were collected. Phosphorylation of IRS1 was used to determine the suitable time for insulin stimulation. For the establishment of the HG-induced IR model, HepG2 cells were cultured with 25 mM of glucose for 0–24 h, and the suitable time for HG induction was determined by detecting the phosphorylation of IRS1 (Ding et al., 2019). To evaluate the therapeutic effect of FFSLD against HG challenge, HepG2 cells were cultured with glucose (25 mM) and treated with FFSLD at the concentrations of 0, 125, 250, 500, 1000, 2000 μg/mL for 24 h, and then stimulated with insulin (1 μM) for 30 min. The protein samples were collected and prepared for Western Blot assay.

2.6. Determination of ROS in HepG2 cells

HepG2 cells were pretreated with or without different concentrations of FFSLD (0, 1000, 2000 μg/mL) for 24 h in high glucose-DMEM (25 mM glucose) medium or low glucose-DMEM (4.5 mM glucose) medium. DCFH-DA was then used to stain the cells according to the instructions of the commercial ROS staining kit. Flow cytometry (Beckman Coulter, California, USA) was used to detect ROS levels of HepG2 cells after FFSLD treatment.

2.7. Animal study

A total of 60 SPF-grade C57BL/6J mice (male, 5–6 weeks, 18 ± 2 g) were purchased from the Experimental Animal Center of Guangdong Province, Guangdong, China (No.44007200067929). All the mice were raised under the SPF condition with 12 h dark-light cycle and had free access to sterile water and diet. The temperature of the environment was 22 ± 2 °C and the humidity was 50 ± 2%. All animal studies were approved by the experimental animal ethics and welfare committee (AEWC) of Zhongshan Hospital of Traditional Chinese Medicine (No. AEWC-016) and performed under the detailed laboratory animal guideline. After one week of adaptive feeding, mice were randomly divided into 6 groups: the control group which was fed with a normal diet, and 5 diabetes-modeling groups which were fed with HFD (consisting of 26.17% Casein, 0.39% L-cystine, 16.35% maltodextrin, 9.00% sucrose, 6.54% cellulose, 3.27% soybean oil, 32.06% lard, 4.58% mineral ain-93, 1.31% vitamin ain-93, 0.33% choline chloride, 1.0% cholesterol, 0.15% bile salt and provided by Guangdong Experimental Animal Center) for 6 weeks, followed by streptozocin injection (STZ, 120 mg/kg, dissolved with citric acid buffer, 0.1M, pH 4.4) after 6 weeks HFD-feeding. Three days later, all mice have fasted for 12 h and the levels of fasting blood glucose (FBG) were detected via a glucose meter (Breeze 2, Bayer). Mice with FBG (tail vein) ≥ 11.1 mmol/L were considered diabetic model mice (Pham et al., 2017). The diabetic mice were divided into 5 groups randomly, including the model, metformin (0.13 g/kg, i.g), the low dose of FFSLD (3.375 g/kg, i.g, calculated according to the extraction yield of crude material, the equivalent dose used in the clinic), the middle dose of FFSLD (6.75 g/kg) and the high dose of FFSLD (13.5 g/kg) groups. The administered low dose of FFSLD was calculated according to the recommended humans dose [175 g/70 kg/day × conversion factor (9) × extraction yield (15%)]. All mice were treated with drugs at the indicated dosages once a day for continuous 12 weeks. The diabetic mice of each treatment group were fed with HFD for 12 weeks continuously, while the control group mice were fed a normal diet. The body weight, water, and food intake of mice were monitored during the experiment, and FBG was measured once a week regularly. After the last treatment, all mice were anesthetized using isoflurane. Blood samples were obtained from the orbital sinus and serum was collected by centrifugation for 15 min at 3000 r/min, 4, and stored at −80 °C until use. The whole animal experimental process is shown in Fig. 2.

2.8. Oral glucose tolerance test (OGTT)

OGTT was performed in the 6 weeks and 12 weeks after STZ injection, respectively. The mice were fasted for 12 h before being administrated with 20% glucose (10 mL/kg, orally). Blood glucose concentrations were measured at 0, 0.5, 1, 2 h after glucose administration by collecting the tail-vein blood. The area under the curve (AUC) of glucose was calculated according to the following formula: AUC = 0.25 × blood glucose (0 h) +0.5 × blood glucose (0.5 h) +0.75 × blood glucose (1.0 h)+0.5 × blood glucose (2.0 h) (Fang et al., 2018).

2.9. Enzyme-linked immunosorbent assay (Elisa)

Elisa assay kits of TC (#TAE-559m), insulin (#TAE-358m), HDL-C (#TAE-334m), LDL (#TAE-407) were obtained from Tianjin Anoric Biotechnology Co., Ltd (Tianjin, China). Elisa assay kits of GC (#MM-0167M2), TG (#MM-1076M2) were purchased from Shanghai Enzyme-linked Biotechnology Co., Ltd (Shanghai, China). The serum biochemical parameters were analyzed following the manufacturer's introductions.

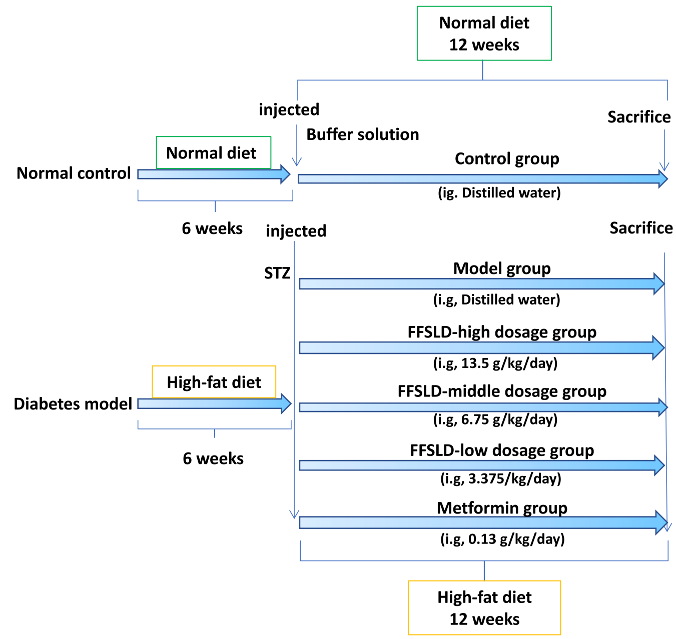

Fig. 2. Design for the anti-T2DM effect of FFSLD in diabetic mice model established by high-fat diet (HFD) feeding combined with streptozotocin (STZ) injection.

2.10. Western Blot

Total cellular protein was extracted using pre-chilled RIPA buffer containing protease and phosphatase inhibitors. An equal amount of protein sample was separated by SDS-PAGE and transferred to polyvinylidene fluoride (PVDF) membranes. The membrane was blocked using 5% non-fat milk in PBS containing 0.1% Tween-20 (PBST) at 25 °C for 2 h. For the protein detection, the membranes were incubated with corresponding primary antibodies overnight at 4 °C, washed with PBST three times, and then incubated with secondary horseradish peroxidase (HRP)-conjugated antibody (1: 10, 000) at 25 °C for 2 h. The membranes were washed three times and protein bands were visualized by using enhanced chemiluminescence (ECL) reagent (Thermo Fisher Scientific) under a chemiluminescence imaging system (Bio-Rad, CA, USA). Image J was used to perform a densitometric analysis of protein expression.

2.11. Network pharmacological analysis

2.11.1. Collection for active compounds of FFSLD

The main chemicals of FFSLD were collected by searching the TCMSP database (https://tcmspw.com/tcmsp.php) and some literature (Ye et al., 2020; Gong et al., 2021; Zhang et al., 2021, 2022; Zhou et al., 2021b). Under the screening conditions of oral-bioavailability (OB) ≥ 30%, drug-likeness (DL) ≥ 0.18, and ADME screening (http://www.swissadme.ch/), the potential active compounds were screened out based on their pharmacokinetics and drug-likeness through TCMSP and Swiss Target Prediction (http://www.swisstargetprediction.ch/) databases. The compound names and molecular structures were checked using the PubChem database (https://pubchem.ncbi.nlm.nih.gov/) and then integrated and deduplicated for collection.

2.11.2. Targets prediction of FFSLD against T2DM

The targets of active compounds were collected based on the prediction and deduplication through TCMSP and Swiss Target Prediction databases. All target names were rectified via the Uniprot database (https://www.uniprot.org/). All genes information was obtained using "Type II diabetes mellitus" as the keyword to search databases of Genecards (https://www.genecards.org/) and OMIM (http://omim.org/).

“Venn” diagram was used to demonstrate the common targets of FFSLD and T2DM.

2.11.3. Network construction

Cytoscape 3.6.1 software (http://www.cytoscape.org/) was used to establish the FFSLD-compound-target-T2DM network, with the parameters being “minimum required interaction score = 0.97″ and “hide disconnected nodes in the network”. The protein-protein interaction (PPI) network was constructed via the String database (https://string-db.org/) by analyzing the common targets with the species limited to *Homo sapiens*.

2.11.4. GO analysis and KEGG pathway enrichment

The predicted target genes were imported into the David v 6.8 database (https://david.ncifcrf.gov/) for enrichment analysis. The major biological processes, cellular components, and molecular functions were analyzed via Gene Ontology (GO; http://www.geneontology.org/) and potential biological pathways were analyzed via Kyoto Encyclopedia of Genes and Genomes platform (KEGG; http://www.kegg.jp/), respectively.

2.12. UHPLC-QTOF-MS analysis

FFSLD powder was dissolved in methanol/water (1:4, v/v) solution at a concentration of 55.2 mg/mL under sonication for 40 min, and then centrifuged at 13,000 rpm for 10 min. The supernatant was collected and filtered with a filter (0.22 μm). Compounds identification was performed through the LC-QTOF-MS method by a C_{18} column (150 mm × 3.0 mm i.d., 1.8 μm; Zorbax). Water containing 0.1% formic acid (solvent system A) and acetonitrile (solvent system B) was served as the mobile phase. The gradient elution program was 0.01–2 min, 5% B; 2–20 min, 5–35% B; 20–24 min, 35–40% B; 24–25 min, 40–50% B; 25–30 min, 50–950% B; 30–35 min, 95% B. The sample injection volume was 0.5 μl and the flow rate was 0.5 ml/min with a column temperature of 30 °C. Mass detection was performed using a QTOF MS 6545 (Agilent, United States) in both positive and negative mode electrospray ionization. The operating parameters were as follows: the ion source gas temperature was 300 °C with a flow rate of 8 L/min; nitrogen was used as a nebulizer and pressure was 45 psi; the sheath gas temperature was 350 °C with a flow rate of 11 L/min; ion spray voltages for positive and negative modes were 4 and −3.5 kV; MS scan range was 100–1000; collision energy (CE): 10/20/40V; MSMS scan range was 50–1000. All data was processed by Analyst software, MassHunter (Agilent).

2.13. Statistical analysis

GraphPad Prism 8 (GraphPad Software, Inc., La Jolla, CA, USA) was used to statistically analyze the data. All results were presented as the mean ± SEM. One-way analysis of variance (ANOVA) and Student’s t-test were used to compare the difference between groups. p-values of <0.05 were considered to be statistically significant.

3. Results

3.1. FFSLD significantly alleviated T2DM on mice

We firstly evaluated the alleviated function of FFSLD *in vivo* in the T2DM mice model induced by HFD feeding combined with STZ injection via assessing the changes of biochemical indexes such as FBG, GC, TC, TG, HDL-C, LDL-C. As shown in Fig. 3A, the mean value of FBG in the control group was 6.3 ± 0.66 mmol/L at the beginning of the experiment, which was normal and stable during the whole process, while the FBG value in the diabetes model group was significantly higher than that of the control group from the beginning to the termination of the experiment, indicating the diabetes model established successfully. After drugs treatment, the FBG values in the FFSLD (low, middle, and high doses) groups and metformin group showed a continuously declined trend compared to the model group from weeks 2–12. Compared with the control group, the FBG level in the model group was significantly increased (##$p < 0.01$). However, after FFSLD and metformin (0.13 g/kg) treatments, the FBG levels were strongly reduced (**$p < 0.01$, compared with the model group) (Fig. 3A). Furthermore, the OGTT showed that the mean value of blood glucose AUC in the model group was much higher (2.8 fold) than that of the control group at week 6 and kept at the same level at week 12, and both FFSLD and metformin down-regulated the blood glucose AUC at week 6 and week 12 (Fig. 3B). The decreased levels of insulin in diabetic mice were significantly increased after FFSLD and metformin treatments (Fig. 3C). Additionally, the levels of other abnormal biochemical indexes including GC (Fig. 3D), TC (Fig. 3E), TG (Fig. 3F), HDL-C (Fig. 3G), LDL-C (Fig. 3H) in diabetic mice were significantly reversed after FFSLD and metformin treatments, indicating the excellent therapeutic efficacy against T2DM.

Altogether, these data validated that FFSLD could significantly alleviate T2MD in the mice model by reducing glucose intolerance, insulin resistance, and dyslipidemia. However, the underlying pharmacological mechanism remains unclear.

3.2. Network pharmacological analysis

3.2.1. Prediction for active ingredients of FFSLD and therapeutic targets against T2DM

After being taken orally, active compounds from herbal medicine usually would undergo the process of absorption, distribution, metabolism, and elimination (ADME). Hence, the parameters of oral-bioavailability (OB) and drug-likeness (DL) are important for the prediction of the active ingredients via network pharmacology. In our study, under the screening condition of OB ≥ 30% and DL ≥ 0.18, 106 candidate active compounds were screened out from FFSLD (Supplementary Tables S1) and 706 targets of the active compounds were picked out via searching the databases of TCMSP and Swiss Target Prediction. In addition, 10103 T2DM-related targets were found through Genecards and OMIM databases. Subsequently, the common targets of FFSLD and T2DM were used to create a Venn diagram (Fig. 4A), in which the overlapping 628 targets (Supplementary Table S2) are representing the potential therapeutic targets of FFSLD against T2DM.

3.2.2. Construction of FFSLD-compound-target-T2DM network

Cytoscape 3.6.1 software was used to construct the FFSLD-compound-target-T2DM network. The candidate active compounds and 628 common target genes of FFSLD-T2DM were introduced into the system for visual processing. The FFSLD-compound-target-T2DM network was constructed by linking candidate compounds to their putative targets. As shown in Fig. 4B, the polychromatic rectangles were representing candidate compounds of FFSLD, and light blue circulars were showing the therapeutic targets of T2DM.

3.2.3. Construction of PPI network

Through importing the “compound–disease-targets” information into the STRING database, the “compound–T2DM target” PPI network with a higher degree (minimum required interaction score = 0.97) of connectivity was constructed, in which 628 nodes and 871 edges were included (Fig. 5A), and the top 30 potential targets of FFSLD against T2DM were shown in Fig. 5B, including AKT1, JUN, EP300, CTNNB1, TNF, CDK1, SRC, CREBBP, HSP90AA1, PTPN11, STAT1, IL6, CCNA2, MYC, ESR1, PIK3R1, VEGFA, CCNB1, CCNB2, CDK2, NCOR2, CDKN1A, CCND1, EGFR, MAPK1, NCOA1, NCOR1, RELA, FN1, IL10.

3.2.4. Enrichment analysis

The R-package clusterProfiler was used to elucidate the biological process (BP), cellular composition (CC), and molecular function (MF) (Supplementary Table S3) of the 628 target proteins. The top 10 BP, CC,

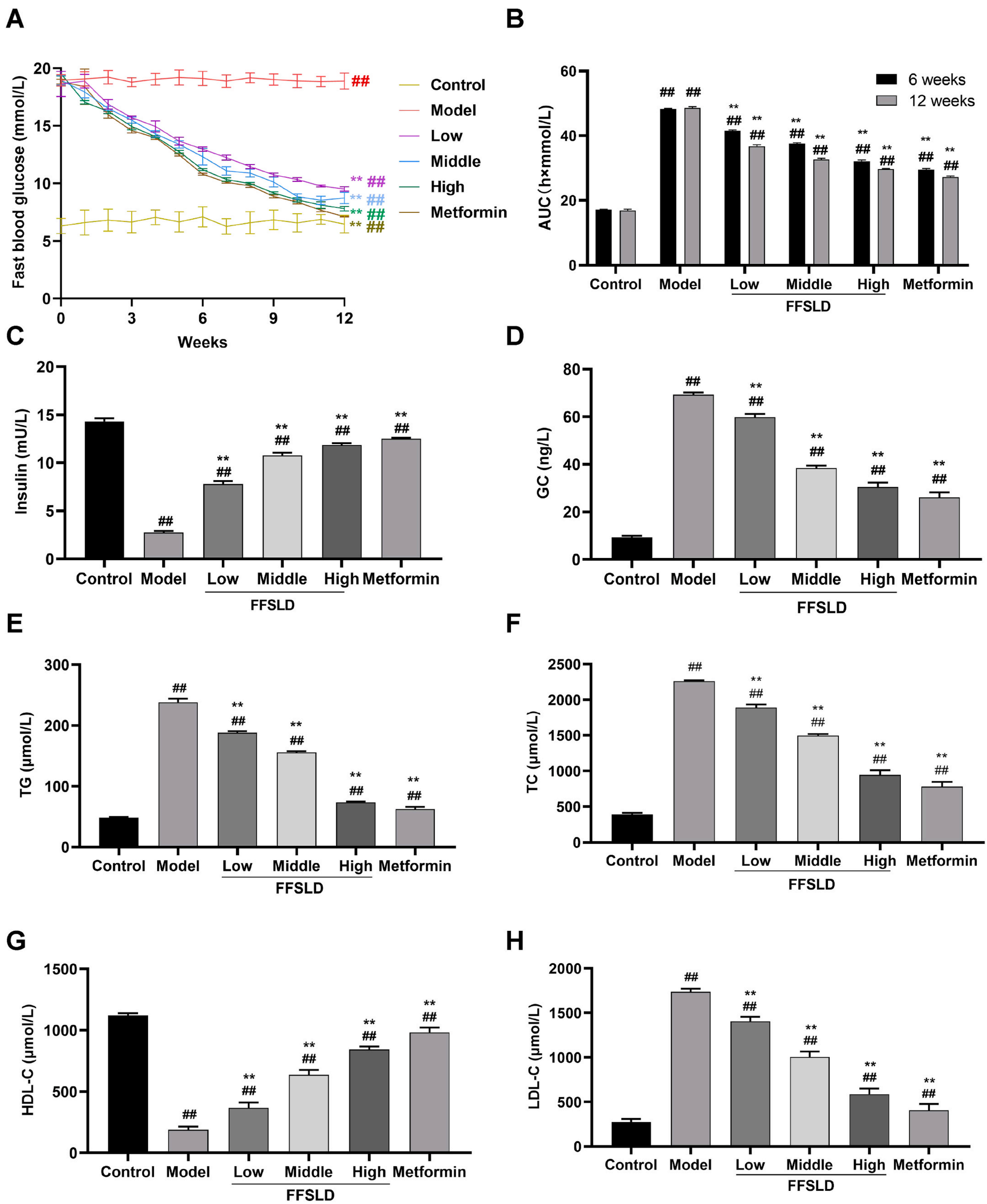

Fig. 3. FFSLD alleviated T2DM by reducing glucose intolerance, insulin resistance, and dyslipidemia in diabetic mice model established by HFD feeding combined with STZ injection. (A) Fast blood glucose levels of the oral glucose tolerance test (OGTT). (B) The area under the curve (AUC) of blood glucose of each group of mice. The levels of insulin (C), glucagon (GC) (D), total cholesterol (TC) (E), triglyceride (TG) (F), high-density lipoprotein cholesterol (HDL-C) (G), and low-density lipoprotein cholesterol (LDL-c) (H) in mice serum were determined via Elisa assay. Data are expressed as mean ± SEM. $^{**}p < 0.01$, compared with the model group; $^{\#\#}p < 0.01$, compared with the control group.

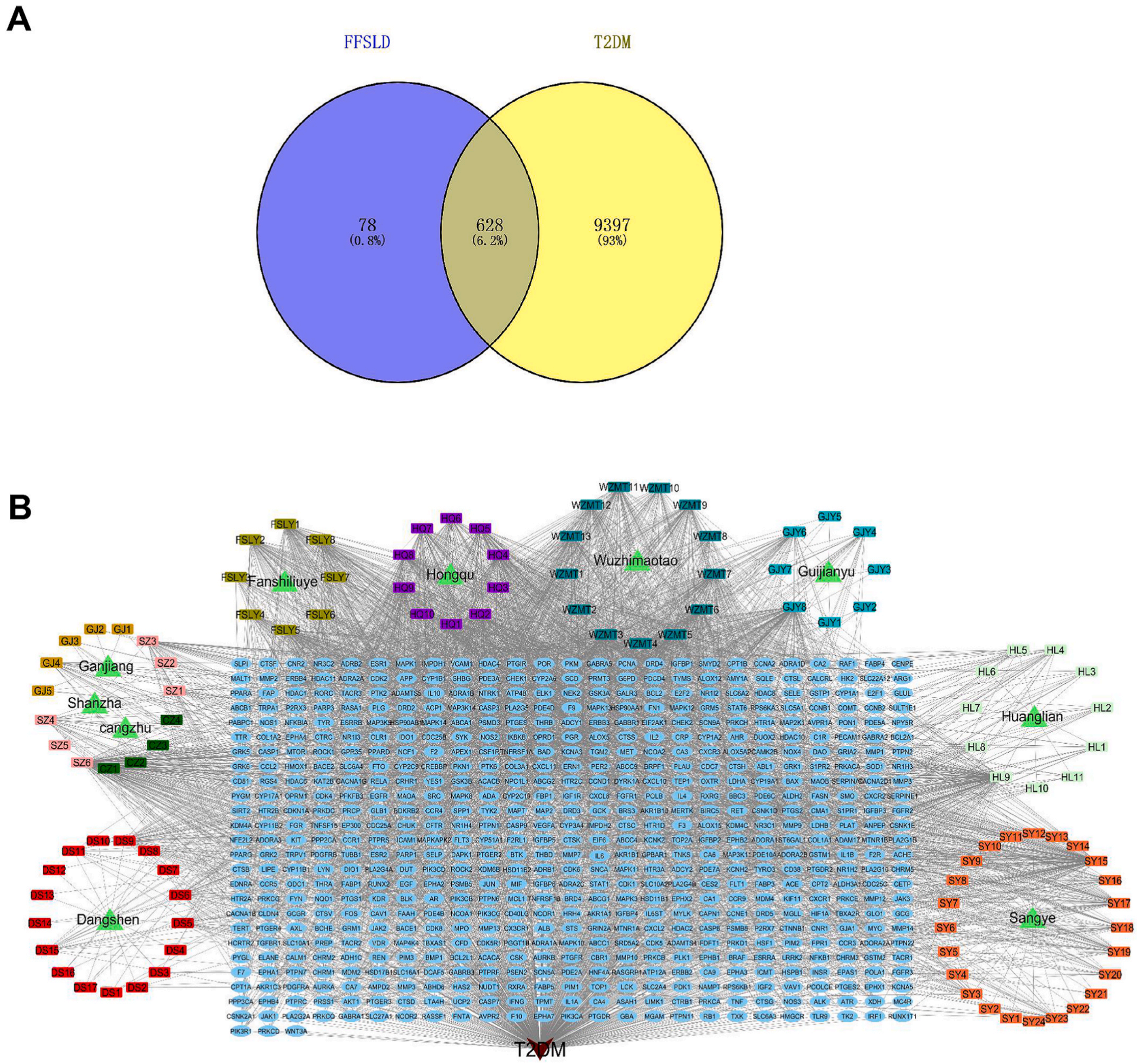

Fig. 4. (A) Venn diagram: overlapping targets between FFSLD components and type II diabetes mellitus (T2DM). (B) Construction of FFSLD-component-target-T2DM network.

MP of the GO analysis data, which were ranked according to their P-values, were shown in Fig. 6A.

Additionally, the KEGG enrichment data showed that 433 signaling pathways in all were included (Supplementary Table S4), and the top 20 pathways of high counts were shown in Fig. 6B. The results showed a high response value to insulin resistance ($P = 5.88844\ E^{-33}$), consistent with our experimental results of FFSLD alleviating T2DM in the mice model. Furthermore, several other functional terms with high response values were also included, such as pathways in cancer, neuroactive ligand-receptor interaction, hepatitis B, prostate cancer, apoptosis, chemokine signaling pathway, cellular senescence, inflammatory mediator regulation of trp channels, progesterone-mediated oocyte maturation, bladder cancer, vascular smooth muscle contraction, NF-kappa B signaling pathway, JAK-STAT signaling pathway, amoebiasis, retrograde endocannabinoid signaling, cGMP-PKG signaling pathway, dopaminergic synapse, cytokine-cytokine receptor interaction, amyotrophic lateral sclerosis (ALS) (Fig. 6B).

3.3. Experimental validation

3.3.1. Establishment of HepG2-IR cells model induced by HG

Insulin receptor substrate 1 (IRS1) is a protein that has multiple phosphorylated sites for receptor kinases, which is a variety of downstream effectors that can be combined (Guo, 2014). Downregulation of p-IRS1 expression is a hallmark during IR occurrence (Ding et al., 2019). In our study, HepG2 cells were stimulated with 1 μM insulin for 0, 5, 15, 30, and 60 min, respectively, and the expression of p-IRS1 in HepG2 cells was detected by Western Blot. As a result, insulin dose-dependently increased the expression of p-IRS1, reaching the peak at 30 min (Fig. 7A and B). Therefore, 30 min of insulin stimulation was chosen in the subsequent experiments. In the HG-induced IR model, HepG2 cells were incubated with 25 mM glucose medium for 0, 3, 6, 12, 18, and 24 h,

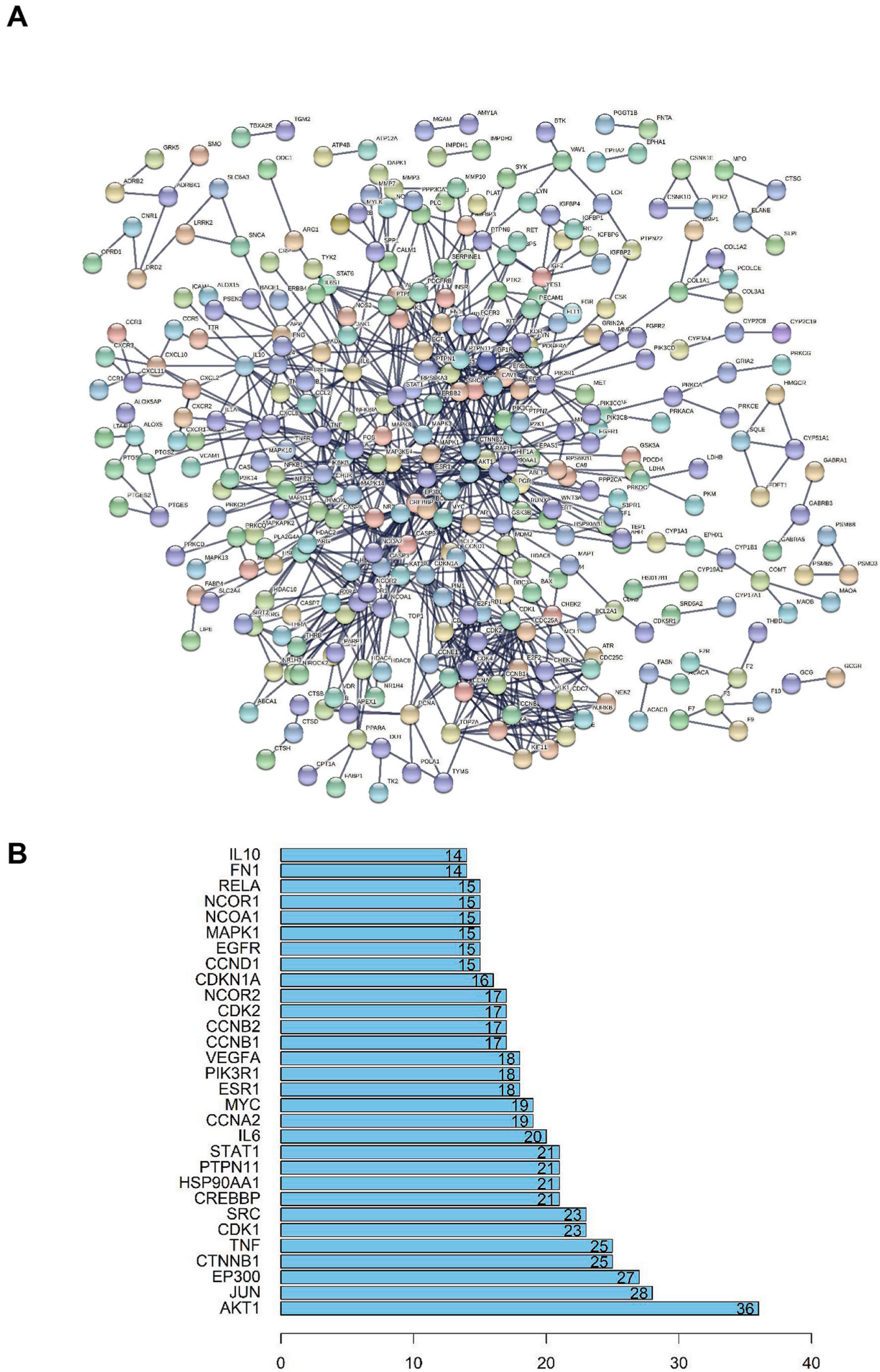

Fig. 5. PPI network. (A) The construction of protein-protein interaction (PPI) network. (B) Bar plot of key nodes from PPI network analysis.

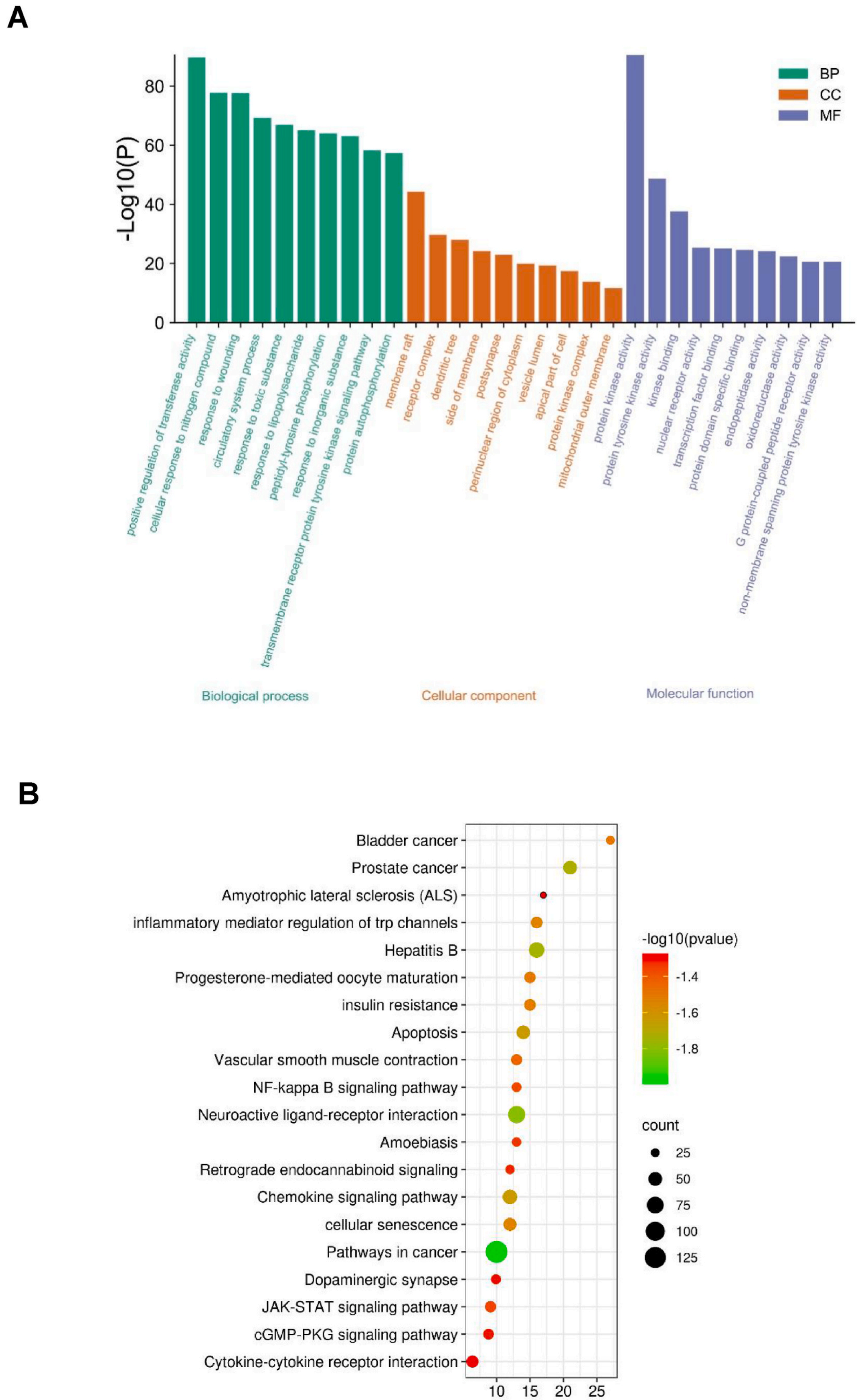

Fig. 6. Gene Ontology (GO) and Koto Encyclopedia of Genes and Genomes (KEGG) enrichment analysis. (A) Bar plot of GO enrichment analysis for FFSLD treating T2DM. (B) Dot plot of KEGG pathway enrichment analysis for FFSLD treating T2DM.

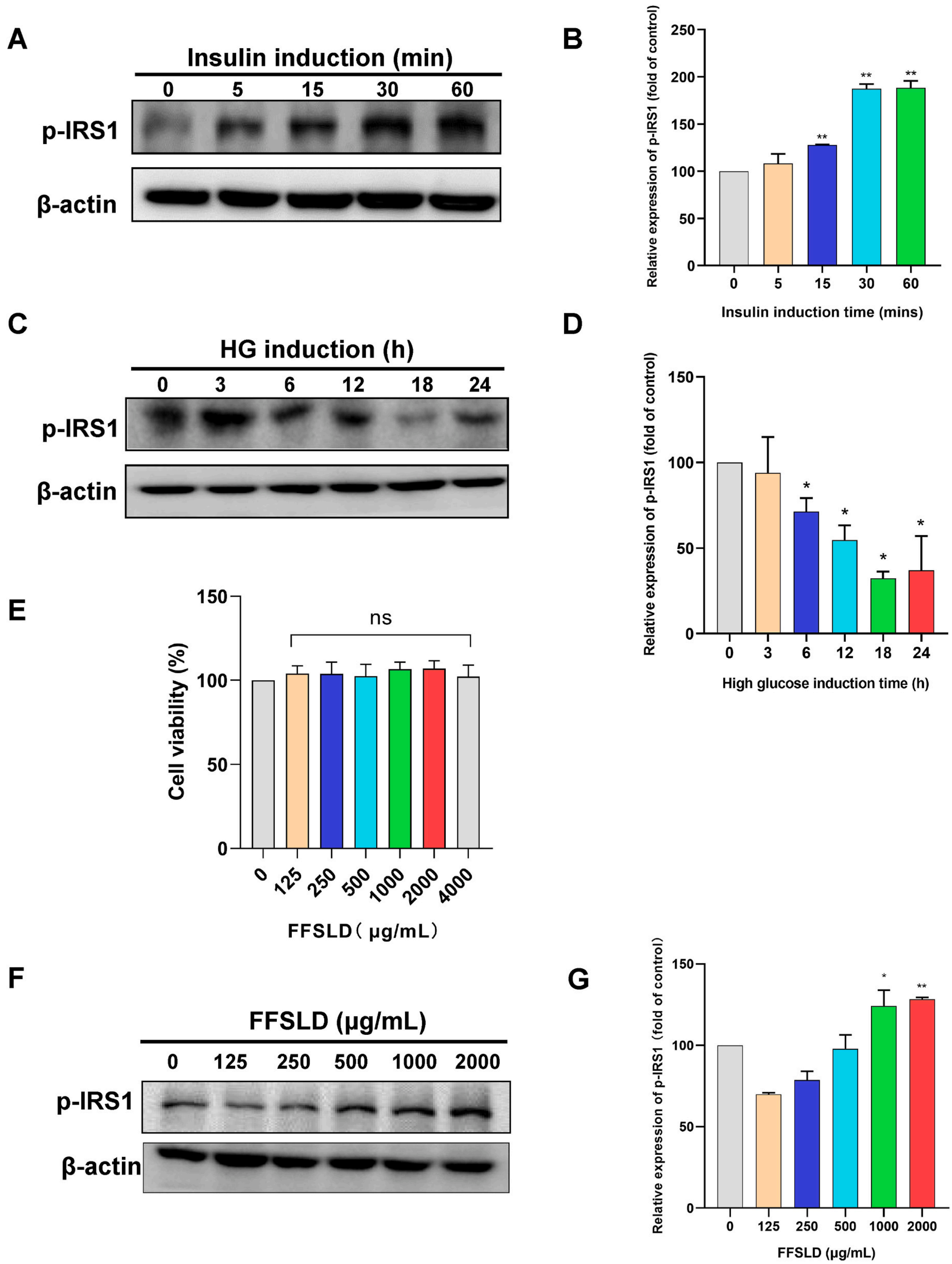

Fig. 7. FFSLD alleviated insulin resistance (IR) in the HepG2-IR cells model induced by high glucose (HG). (A) Western Blot detection and (B) quantitative analysis for the expression of p-IRS1 in HepG2 cells stimulated by insulin (1 μM) for 0, 5, 15, 30, 60 min, respectively. (C) Western Blot detection and (D) quantitative analysis for the expression of p-IRS1 in HepG2 cells incubated with 30 mM glucose for 0, 3, 6, 12, 18, 24 h, respectively, and followed by insulin stimulation (1 μM) for 30 min. (E) Cell viability of HepG2 cells treated by different concentrations of FFSLD for 24 h. (F) Western Blot detection and (G) quantitative analysis for the expression of p-IRS1 in HepG2 cells treated with different concentrations of FFSLD in 30 mM glucose of medium, followed by insulin stimulation (1 μM) for 30 min. Data (n = 3) are expressed as mean ± SEM. **p* < 0.05, ***p* < 0.01, compared with the control group.

respectively, and then treated with insulin (1 μM) for 30 min. The expression of p-IRS1 was detected via Western Blot. As shown in Fig. 7C and D, p-IRS1 was down-regulated in a time-dependent manner, reaching the lowest level at the 24 h time-point. Therefore, HepG2 cells treated with 25 mM glucose for 24 h were selected to establish an IR model *in vitro*.

3.3.2. FFSLD alleviated IR in HepG2-IR cells model

To verify the cytotoxicity of FFSLD on HepG2 cells, different concentrations of FFSLD (0, 125, 250, 500, 1000, 2000, 4000 μg/mL) were used to treat HepG2 cells for 24 h, and the cell survival rate was determined by MTT assay. The results showed that cell survival rates were not influenced by FFSLD, without significant difference compared with the control ($p > 0.05$), indicating that FFSLD had little cytotoxicity on HepG2 cells at the indicated concentrations (Fig. 7E).

Subsequently, the ameliorative effect of FFSLD on the HepG2 IR model induced by HG was determined by detecting the expression of p-IRS1. As a result, the p-IRS1 level was significantly up-regulated in a good dose-dependent manner after the treatment of FFSLD (Fig. 7F and G), showing that FFSLD had the capability of alleviating IR in the HepG2 IR model induced by HG. These data indicated the good therapeutic function of FFSLD against T2DM via alleviating IR, as predicted by the network pharmacology analysis. However, the detailed mechanism and potential regulating signal pathways are not clear yet.

3.3.3. FFSLD alleviated IR through regulating PI3K-AKT and ESR1 signaling pathways

It has been reported that Chinese medicines possessing the therapeutic effect against T2DM can alleviate insulin resistance by regulating PI3K/Akt pathway (Lyu et al., 2019). The lipid metabolism was associated with ESR alpha promoter methylation and the risk of T2DM (Zhou et al., 2021a). The PPI network analysis in this study had shown that AKT1 and ESR1 possessed high response values in the T2DM treatment by FFSLD. Our subsequent study verified that p-Akt and ESR1 in the HepG2-IR cells model were significantly up-regulated after FFSLD treatment (Fig. 8A and D and) detected by Western Blot assay, suggesting that FFSLD could regulate PI3K-AKT and ESR1 signaling pathways to alleviate IR for T2DM therapy.

3.3.4. FFSLD alleviated IR by suppressing oxidative stress

In the GO analysis, the MF result showed that the mechanism of FFSLD treating T2DM was probably related to regulating the oxidoreductase activity. Oxidoreductase is involved in many biological processes, especially in the evolution of oxidative stress. Oxidative stress is not only the inducer of T2DM (Tangvarasittichai, 2015) but also the main cause of complications resulting from T2DM (Charlton et al., 2020; Pasupuleti et al., 2020). To testify the oxidative-stress regulation of FFSLD in treating T2DM, flow cytometry was used to determine the effect of FFSLD on the ROS level of HepG2-IR cells. The results showed that FFSLD significantly reduced the ROS level of HepG2-IR cells induced by HG, close to that of the LG group (Fig. 9A and B). In addition, the expressions of Nrf2 and Keap1, two key factors of the oxidative stress pathway, were evaluated via Western Blot assay. As shown in Fig. 9C and E, FFSLD significantly up-regulated the expression of Nrf2 and down-regulated the expression of Keap1, suggesting that FFSLD could reduce the oxidative stress level of the HepG2-IR cells model by regulating the Nrf2-ARE pathway, thus promoting the efficacy in T2DM treatment.

3.3.5. FFSLD alleviated IR by suppressing JAK2/STAT3 pathway

The JAK/STAT signaling pathway, a kind of typical pro-inflammatory pathway, plays an important role in the occurrence and development of T2DM, especially in the biological process of lipid metabolism and insulin resistance (Dodington et al., 2018). In our study, the network pharmacology results showed that JAK/STAT pathway had a high response value in the process of FFSLD treating T2DM. Western Blot assay verified that the expressions of JAK2 and STAT3 were significantly down-regulated in HepG2 IR cells under the challenge of FFSLD (Fig. 9F and H), suggesting that FFSLD could regulate the JAK/STAT pathway to treat T2DM.

3.4. Compounds identification of FFSLD determined by UHPLC-QTOF-MS

Accurate mass data were acquired in the full-scan analysis, and production mass data were acquired in MS/MS analysis. As a result, a total of 457 compounds (Supplementary Table S5), including securinol C, adenosine, 6,7-Dihydroxy-1,1-dimethyl-N-(2′-glyceryl)-1,2,3,4-tetrahydroisoquinoline, kynurenine, 7-Hydroxycoumarin, bujeine, 6-

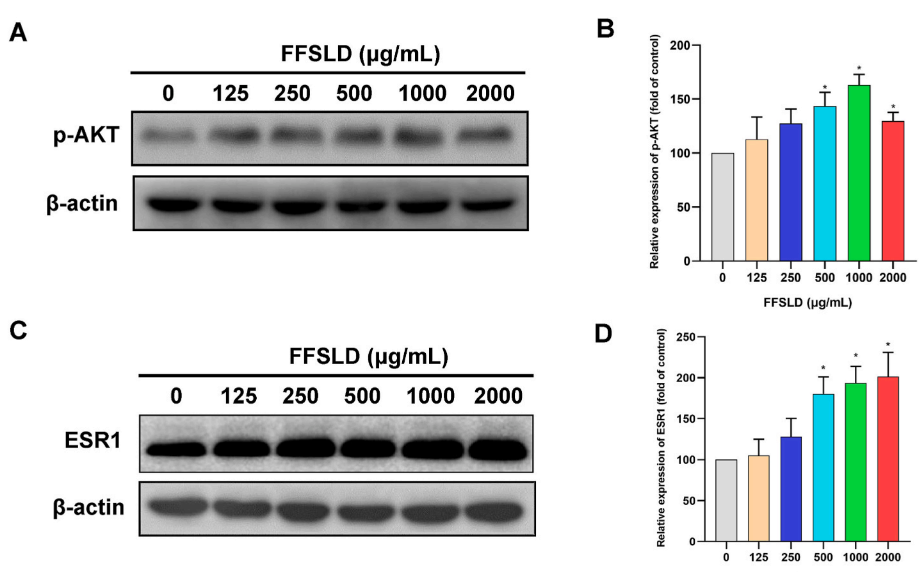

Fig. 8. FFSLD alleviated IR in the HepG2-IR cells model by regulating the PI3K-AKT and ESR pathways. (A, C) Western Blot detection and (B, D) quantitative analysis for the expressions of p-AKT, ESR1 in HepG2-IR cells treated with different concentrations of FFSLD for 24 h. Data (n = 3) are expressed as mean ± SEM. *$p < 0.05$, comparing with control.

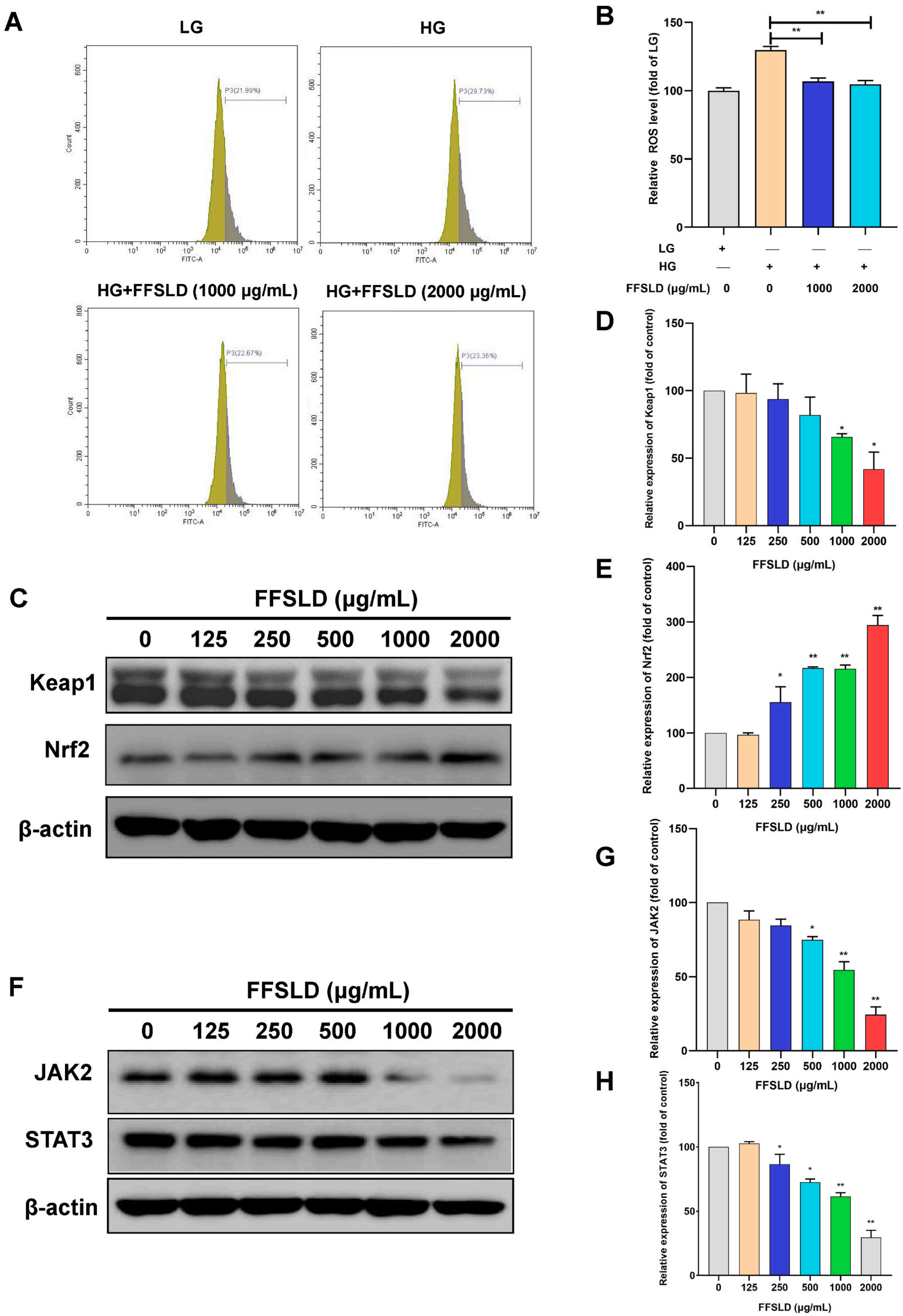

Fig. 9. FFSLD suppressed the oxidative stress and JAK/STAT pathways in the HepG2-IR cells model. (A) Flow cytometry assay and (B) quantitative analysis for ROS levels of HepG2 cells induced by low glucose, high glucose, high glucose + FFSLD. (C and F) Western Blot detection and (D, E, G, and H) quantitative analysis for the expressions of Keap1, Nrf-2, JAK2, and STAT3. Data (n = 3) are expressed as mean ± SEM. **p* < 0.05, ***p* < 0.01, compared with the model group (without FFSLD treatment).

Methylcodeine, berberrubine, quercetin-3-O-arabinoside, coptisine, berberine, palmidin A, coptisine chloride, hesperetin, kaempferol, quercetin-3-O-(2''-O-alpha-rhamnosyl-6''-O-malonyl)-beta-glucoside et al., were identified in the positive and negative modes (Fig. 10).

4. Discussion

As a kind of harmful disease, T2DM causes severe medical resources consumption of society. Meanwhile, many complications associated with T2DM bring mental and physical torture to diabetic patients (Mezuk et al., 2008; Lu et al., 2017). According to the data from International Diabetes Federation Diabetes Atlas, there were 4.2 million aged 20-79 adults dead from diabetes in 2019 globally, which equaled to eight death every minute (Saeedi et al., 2020). Current therapeutic drugs used in the clinic such as metformin could effectively control the patients' blood glucose. Other complementary remedies for diabetes are still needed to meet the needs of such a huge number of patients. Traditional ethnic medicine has a long history of clinical application and takes effect through multiple channels and multiple targets, and would be considered to be a potential complementary remedy for T2DM treatment.

IR refers to a state in which the target tissues of insulin, such as the liver, skeletal muscle, and adipose tissue, become insensitive to insulin, and is the main cause of T2DM (Tangvarasittichai, 2015). T2DM would also interact with other diseases, such as hypertension, obesity, and hyperlipidemia (Zheng et al., 2018). In 1993, Reaven named this group of diseases "metabolic syndrome X" (Reaven, 1993). In 1995, Stern proposed the "common soil theory", which indicated that the common soil was IR, in which these diseases would grow (Stern, 1995). IR is regarded as a "silent" risk factor, and an important cause of T2DM or other metabolic diseases (Samuel and Shulman, 2012). It is reported that IR can be induced by free fatty acids (Geng et al., 2017; Mo et al., 2019) and HG (Ding et al., 2019) by inhibiting the p-IRS1 expression and reducing glucose consumption in HepG2 cells. Thus, increasing p-IRS1 expression in the IR model implies a therapeutic effect on T2DM.

FFSLD, a kind of TCM decoction, has been used in the clinic for many years, with good efficacy in treating diabetes. According to the basic theory of Chinese medicine of "*Jun-Chen-Zuo-Shi*", *Psidium guajava* L. (Fanshiliuye) is the "Jun" drug among FFSLD, which has been recorded to treat diabetes in the ethnic medicinal book (Chen and Zhang, 1995). Additionally, other herbal medicines such as *Morus alba* L. (Sangye) (Lyu et al., 2021), *Euonymus alatus* (Thunb.) Sied. (Guijianyu) (Dong, 2018), *Coptis Chinensis* (Huanglian) (Di et al., 2021), *Crataegus pinnatifida* B. (Shanzha) (Liu et al., 2021), and *Monascus purpureus* W. (Hognqu) (Hu et al., 2020) had been reported to possess the anti-diabetes effect and showed beneficial effects in prevention of metabolic diseases related to T2DM through various pathways. Our previous clinical study showed that a combination of FFSLD and metformin significantly improved the pathological indicators of WC, BMI, FBG, P2hBG, HbA1c, TC, TG, FFA, LDL-C, HDL-C, HOMA-β in T2DM patients with super-heavy obesity or metabolic syndrome compared with the metformin-only-treatment group (Cheng et al., 2016; Li et al., 2016). FFSLD was also determined to inhibit the activities of α-amylase and α-glucosidase (Dai et al., 2016). Additionally, FFSLD decreased the levels of blood glucose and lipid and protected pancreatic tissue in T2DM rats (Li et al., 2015). According to the American Diabetes Association guidelines for diabetes diagnosis and treatment, people with FBG $\geq$7 mmol/L and 2 h postprandial glucose (2hPG) $\geq$ 11.1 mmol/L can be diagnosed as diabetes mellitus patients, and the levels of GC and insulin, two important factors reflecting the FBG level, are used to estimate the pancreatic islet cell function. A low level of insulin combined with a high level of FBG and GC are commonly indicating IR in the T2DM model. In our current study, the therapeutic efficacy of FFSLD against diabetes has been verified in the T2DM mice via decreasing the FBG level and recovering the abnormal biochemical indexes, including insulin, GC, TC, TG, HDL-C, LDL-C. Combining with the OGTT data detected in week 6 and week 12, it is concluded that FFSLD significantly improves the glucose tolerance and IR in the T2DM mice model. Furthermore, the up-regulation of p-IRS1 in HepG2-IR cells implied the ameliorated effect of FFSLD against IR. Patients with long-term high FBG are always accompanied by abnormal lipid metabolisms, such as high levels of TG, TC, LDL-C, and low levels of HDL-C in T2DM (Jafari-Maskouni et al., 2020). Our study showed that FFSLD

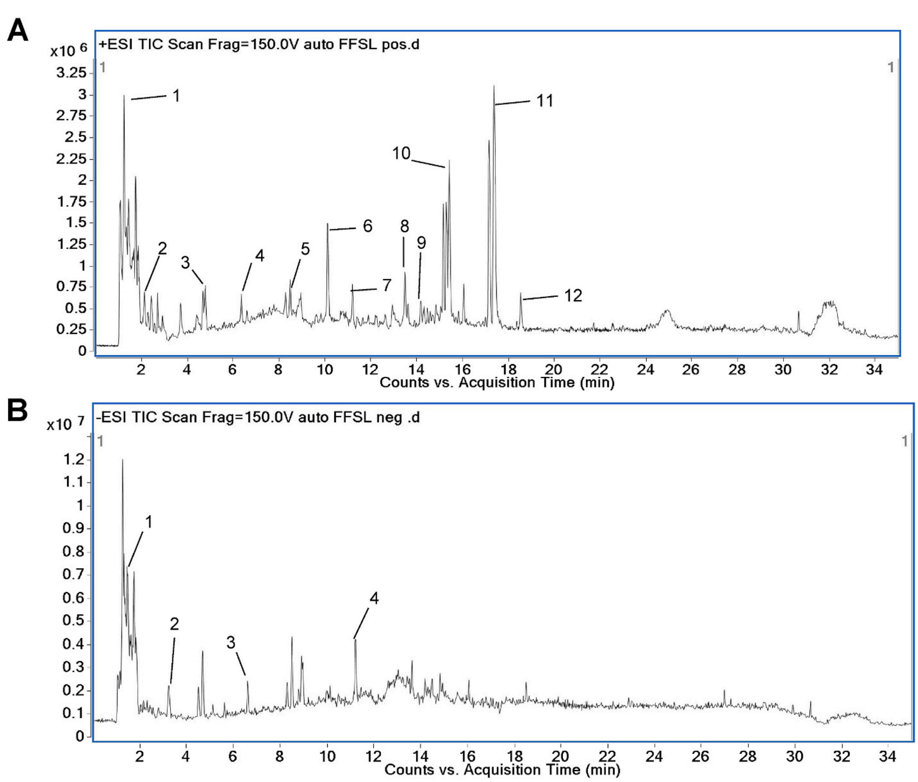

Fig. 10. Representative base peak chromatogram of FFSLD in the positive and negative ions mode, respectively. (A) Positive ions mode: (1) securinol C, (2) adenosine, (3) 6,7-Dihydroxy-1,1-dimethyl-N-(2′-glyceryl)-1,2,3,4-tetrahydroisoquinoline, (4) kynurenine, (5) 7-Hydroxycoumarin, (6) bujeine, (7) 6-methylcodeine, (8) berberrubine, (9) quercetin-3-O-arabinoside, (10) coptisine, (11) berberine, (12) palmidin A; (B) negative ions mode: (1) coptisine chloride, (2) hesperetin, (3) kaempferol, (4) quercetin-3-O-(2″-O-alpha-rhamnosyl-6″-O-malonyl)-beta-glucoside.

significantly recovered the abnormal lipid metabolism by decreasing the levels of TG, TC, LDL-C, and increasing the level of HDL-C. In all, these data suggested that FFSLD improved the glucose tolerance, IR, and lipid metabolism of T2MD model mice induced by HFD feeding combining STZ.

Network pharmacology is commonly utilized in the exploration of drug mechanisms, especially for some drugs with complex components. Thus, it showed a great advantage in using network pharmacological analysis to explore the pharmacological mechanisms of FFSLD against T2DM. In this study, a multidimensional regulatory network comprising "FFSLD-compound-target-T2DM" was established via network pharmacology to identify the potential pharmacodynamical substances in FFSLD and elucidate their pharmacological mechanism in alleviating IR. The results showed that 106 active compounds of FFSLD including 17 compounds identified in the UHPLC-QTOF-MS analysis (Figs. 10), and 628 therapeutical targets were predicted by analyzing the targets of active compounds from FFSLD and T2DM therapeutic targets. In GO analysis (Supplementary Table S3), a total of 4492 items, including 3797 biological processes, 228 cellular compounds, and 467 molecular functions, were enriched for analysis. In KEGG analysis, the total 433 signaling pathways (Supplementary Table S4) were enriched for analysis, and a high response value to insulin resistance ($P = 5.88844\ E^{-33}$) was predicted, supporting the animal study and clinical usage of FFSLD against T2DM. Furthermore, experimental evaluation was performed to validate the mechanism of FFSLD on the "high response" pathways in IR alleviation, including PI3K/AKT, ESR, oxidative stress, and JAK/STAT pathways.

PI3K/Akt pathway is the classical pathway of insulin signaling, of which the key protein Akt regulates glucose metabolism by phosphorylating multiple downstream target proteins and regulating transcription factors (Jiang and Zhang, 2002). p-AKT was up-regulated after FFSLD treatment showing the alleviation of IR. ESR1 has been reported to be closely related to the occurrence and development of T2DM (Yang et al., 2018), and serves as a biomarker for the early diagnosis of T2DM (Linnér et al., 2013). Gegen Qinlian decoction, another Chinese herbal medicine, was reported to markedly decrease blood glucose and increase serum insulin levels in type II diabetic mice via increasing ESR1 expression (Xu et al., 2020). The promotion of ESR1 expression was also validated in the FFSLD-treated HepG2 IR cells model. Oxidative stress plays a critical role in the pathogenesis of IR through destructing insulin signals (Hurrle and Hsu, 2017). Studies showed that the depletion of Nrf2, a key protein of the anti-oxidative stress pathway, could increase blood glucose levels, aggravate glucose intolerance, and impair insulin signaling (Hayes and Dinkova-Kostova, 2014). Correspondingly, our study verified that FFSLD alleviated oxidative stress and IR by modulating the Keap1-Nrf2 signaling pathway. JAK/STAT signaling pathway plays an important role in the process of adipogenesis, IR (Mishra et al., 2015; Dodington et al., 2018), and triggering of T2DM. It has been reported that alterations of muscle mass and skeletal muscle IR are probably the early initiating events in T2DM, and STAT3 silencing could prevent the development of IR induced by lipid in rat myotubes (Mashili et al., 2013). Our study showed that the expressions of JAK2 and STAT3 were down-regulated indicating the suppression of FFSLD in the HG-induced HepG2 IR model.

Altogether, FFSLD significantly improved HFD/STZ-induced T2DM *in vivo* by reversing the abnormal levels of FBG, insulin, GC, TC, TG, HDL-C, LDL-C in mice. Mechanistically, FFSLD alleviated IR by increasing the expression of p-IRS1 in the HepG2-IR cells model, increasing the expressions of ESR1, p-AKT, Nrf-2, and reducing the expressions of JAK2, STAT3, Keap-1, suggesting that FFSLD could regulate PI3K/AKT, ESR, JAK/STAT, and oxidative stress pathways to alleviate IR in HG-induced HepG2 IR model, which was consistent with the prediction of network pharmacological analysis.

5. Conclusion

Our study concluded that FFSLD could significantly alleviate T2DM disease by reducing insulin resistance. The underlying mechanism was clarified to be related to the regulation of PI3K/AKT, ESR, oxidative stress, and JAK/STAT signaling pathways by integrating network pharmacological analysis and experimental validation. Our work provides pharmacological evidence to support the ethnic usage of FFSLD in treating T2DM.

Data availability

The raw data supporting the conclusions of this manuscript will be made available by the authors, without undue reservation, to any qualified researcher.

CRediT authorship contribution statement

Weibo Dai: Investigation, Methodology, Software. **Chang Chen:** Writing – original draft, Investigation, Methodology, Software. **Gengting Dong:** Writing – review & editing, Writing – original draft, Investigation, Methodology, Software. **Guangru Li:** Software, Validation. **Weiwen Peng:** Conceptualization, Supervision, Funding acquisition, Resources. **Xin Liu:** Software, Validation. **Jing Yang:** Software, Validation. **Leyu Li:** Conceptualization, Supervision, Funding acquisition, Resources. **Ruiyan Xu:** Conceptualization, Supervision, Funding acquisition, Resources. **Xianjing Hu:** Writing – review & editing, Writing – original draft, Conceptualization, Supervision, Funding acquisition, Resources.

Declaration of competing interest

The authors declare that there are no conflicts of interest.

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Abbreviations

ANOVA	one-way analysis of variance
BSA	bovine serum albumin
BP	biological process
CC	cell composition
MF	molecular function
DCFH-DA	2,7-Dichlorodi-hydrofluorescein diacetate
DMEM	Dulbecco's Modified Eagle's medium
DMSO	Dimethyl sulfoxide
DL:	drug-likeness
ECL:	enhanced chemiluminescence
ESR1	estrogen receptor alpha
FBG	fasting blood glucose
2hPG	2-h post-load plasma glucose
FBS	fetal bovine serum
FFSLD	Fufang FanShiLiu decoction
GC	glucagon
GO	Gene Ontology
HDL-C:	high-density lipoprotein cholesterol
HFD	high fat diet; HG: high glucose
HRP	horseradish peroxidase
IDF	International Diabetes Federation

IR	insulin resistance
KEGG	Koto Encyclopedia of Genes and Genomes
LDL-C:	low-density lipoprotein cholesterol
MTT	3-[4, 5-dimethylthiazol-2-yl]-2, 5-diphenyltetrazolium bromide
OB	oral-bioavailability
OGTT	oral glucose tolerance test
PBS	phosphate buffered saline
PBST	PBS with 0.1% Tween-20
p-IRS1	phosphorylated insulin receptor substrate 1
PPI	protein to protein interaction network
PVDF	polyvinylidene fluoride
ROS	reactive oxygen species
SDS-PAGE	sodium dodecyl sulfate-polyacrylamide gel electrophoresis
STZ	streptozotocin
TC	total cholesterol
TCM	traditional Chinese medicine
TCMSP	Traditional Chinese Medicine Systems Pharmacology Database and Analysis Platform
T2DM	type II diabetes mellitus
TG	triglyceride; Uniprot: Universal Protein
WebGestalt	web-based gene set analysis toolkit
PKC	protein kinase C
MAPKs	mitogen-activated protein kinases
AMPK	Adenosine monophosphate (AMP)-activated protein kinase
NF-κB	nuclear factor kappa B
2hPG	2 h postprandial glucose

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jep.2022.115338.

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